Data and Intelligence

- Hypothetical scenario
  - Suppose you are given the choice of three doors
    - Behind each door, there is an unknown amount of cash
      - The distributions of the cash behind the three doors are independent
    - You choose a particular door to win the cash behind it
- Goal
  - To maximize the return (win the cash as much as possible)
    - Random picking ➔ ⅓ chance of success
    - Deterministic picking ➔ ⅓ chance of success
- Observation
  - In terms of success, humans cannot outperform monkeys
- Questions
  - Can we do better?
  - Is there a more intelligent approach?

Data and Intelligence

- Introducing additional conditions
  - Collecting data
    - You can open all three doors to count the cash behind them ➔ Trivial
    - You can open any one door to count the cash behind it ➔ Not helpful
    - You can open any two doors to count the cash behind them

Data and Intelligence

- An "intelligent" algorithm
  1. Randomly choose Door #1 and Door #2
  2. Open Door #1, count the cash
  3. Open Door #2, count the cash
  4. If the cash behind Door #1 > the cash behind Door #2, then Door #3 is your final decision
  5. Otherwise, Door #2 is your final decision
- The chance of success (i.e., maximizing the return) increases from **⅓** to **½**
- On average, this algorithm is able to win **(3A-B)/12** more cash than random picking or deterministic picking
  - A = the largest amount of cash behind any of the three doors
  - B = the sum of the cash behind all three doors

Data and Intelligence

- Remarks
  - It is possible to utilize data to maximize the return
  - More data does not mean more return
  - We need intelligence to utilize data to maximize the return

Data + Intelligence ➔ Business Intelligence

Potential Risks

- There are many potential risks that an organization faces when attempting to use business intelligence to outpace the competitors
- Executives should be aware of the following potential risks in advance
  - Awakening a sleeping giant
    - A larger competitor with deeper pockets may be nudged into implementing BI with more data and better technologies
  - Running afoul of the law
    - All BI systems have data security and privacy issues
    - There is no perfect system
  - Attempting to collect “Big Data”
    - The data collected by a private organization can never be “big”
  - Collecting poor quality data
    - Biased data collection process or biased data will result in discriminatory decisions
    - Example - racist AI or sexist AI for automatic loan approval

Potential Risks

- Wrongly interpreting the results for decision making
  - Currently, most data mining and machine learning discoveries are not suitable for long-term decisions
  - Example – the beer and diapers story
    - Many customers bought both beer and diapers at the same time
      - Placing beer and diaper together in the store
      - Separating beer and diaper far away in the store
      - Re-pricing beer and diaper to gain more profit
      - Package beer and diapers as a single bundle
- Using improper analytics methods
  - Improper analytics methods will result in false discoveries
  - Examples
    - Eating ice cream causes drownings
    - Package coffee and tea as a single bundle
    - Simpson's paradox

Data, Information and Knowledge

- Data
  - Raw facts
  - Simple observations
  - Sequences of digits
- Information
  - Data + semantic meanings
  - Relevance
  - Purpose
- Knowledge
  - Information that can guide actions
  - Information + context + synthesis + reflection
- Data, Information and Knowledge are valuable assets

Data, Information and Knowledge

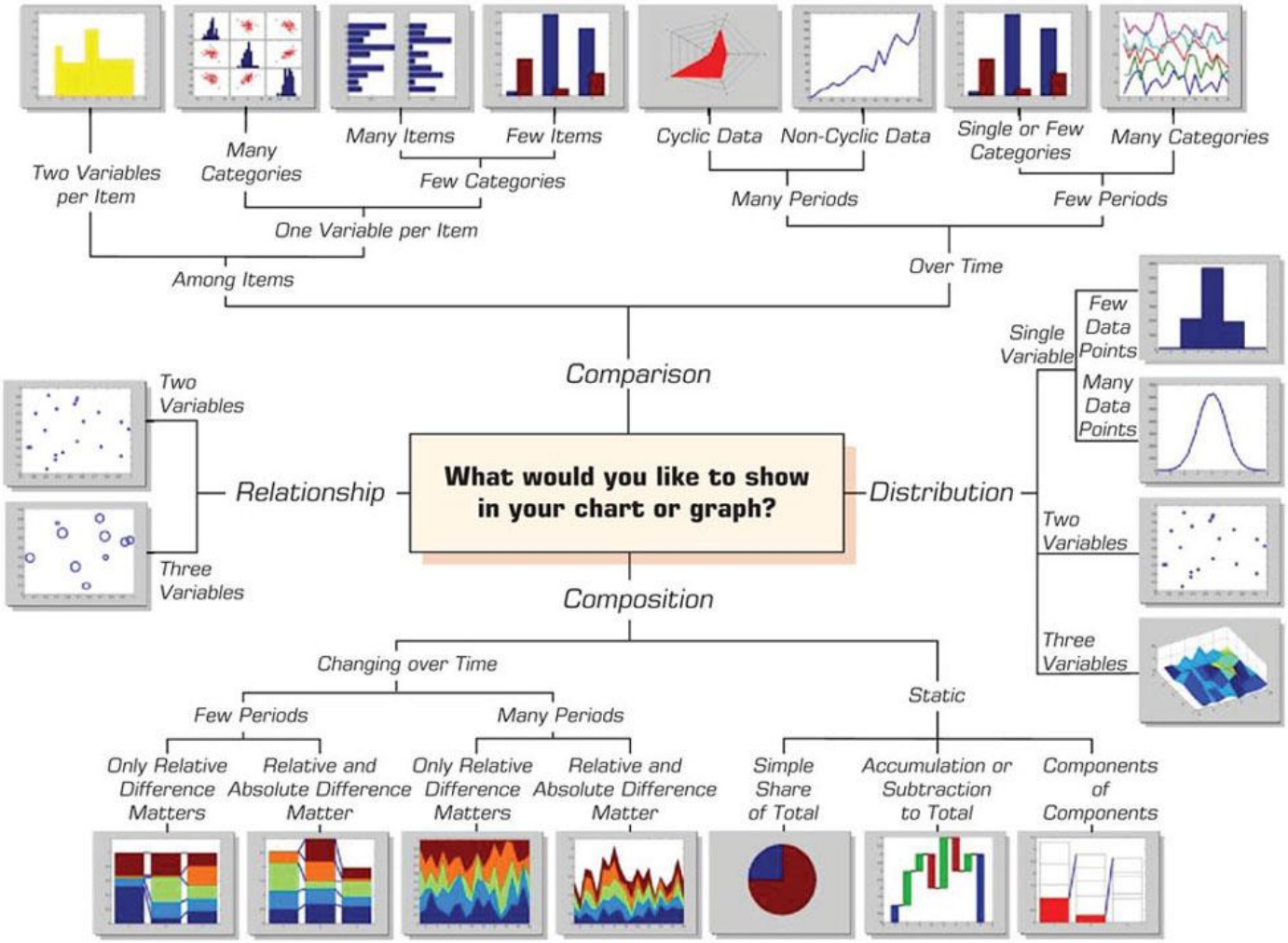
	Data	Information	Knowledge
Definition	Simple observations of the state of the world	Data endowed with relevance and purpose	Information from the human mind (includes reflection, synthesis, context)
Characteristics	• Easily structured • Easily captured on machines • Often quantified • Easily transferred • Mere facts	• Requires unit of analysis • Data that have been processed • Human mediation necessary	• Hard to structure • Difficult to capture on machines • Often tacit • Hard to transfer
Example	Daily inventory report of all items	Daily inventory report of items with low levels of stock	Report of which items need to be reordered after considering inventory levels, anticipated labor strikes, and a flood affecting a supplier

Data and Information Visualization

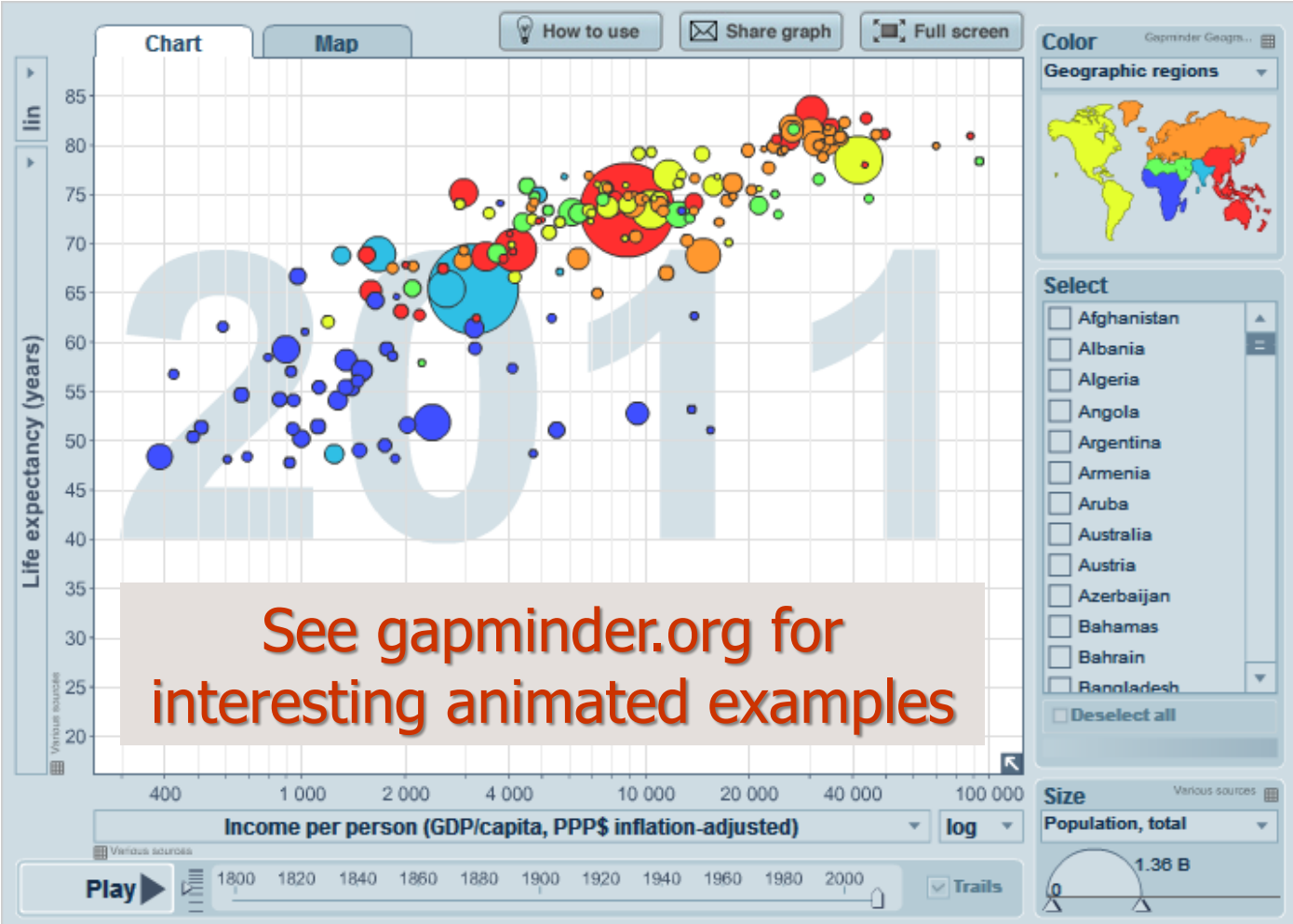
"The use of visual representations to explore, make sense of, and communicate data."

- Basic Charts and Graphs
  - Line Chart
  - Bar Chart
  - Pie Chart
  - Scatter Plot
  - Bubble Chart
- Specialized Charts and Graphs
  - Histogram
  - Gantt Chart
  - PERT Chart
  - Geographic Map
  - Bullet Graph
  - Heat Map / Tree Map
  - Highlight Table

Selection Decisions of Charts and Graphs



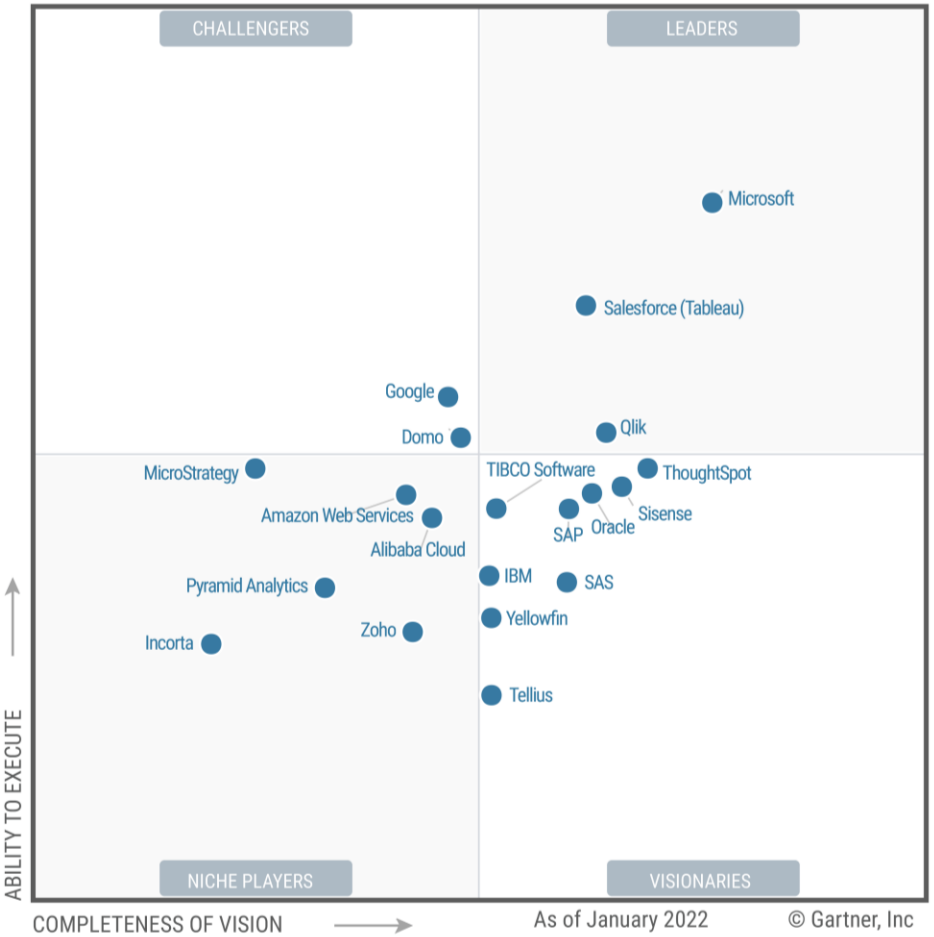
Example - Wealth and Health of Nations



See gapminder.org for interesting animated examples

Example – Magic Quadrant of BI Software

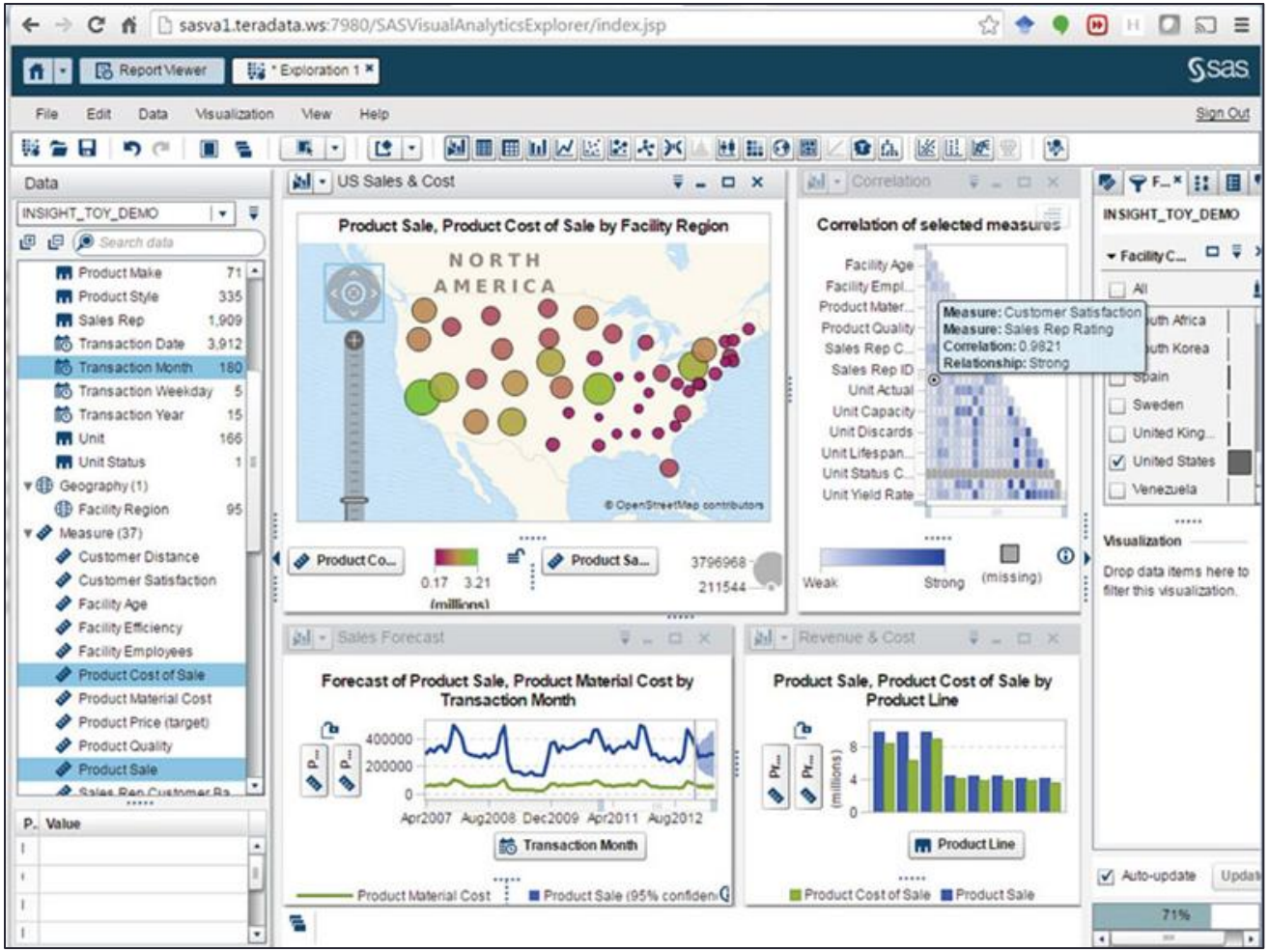
Leading BI software are in the 4th quadrant as of the end of 2022 (Source: Gartner.com)



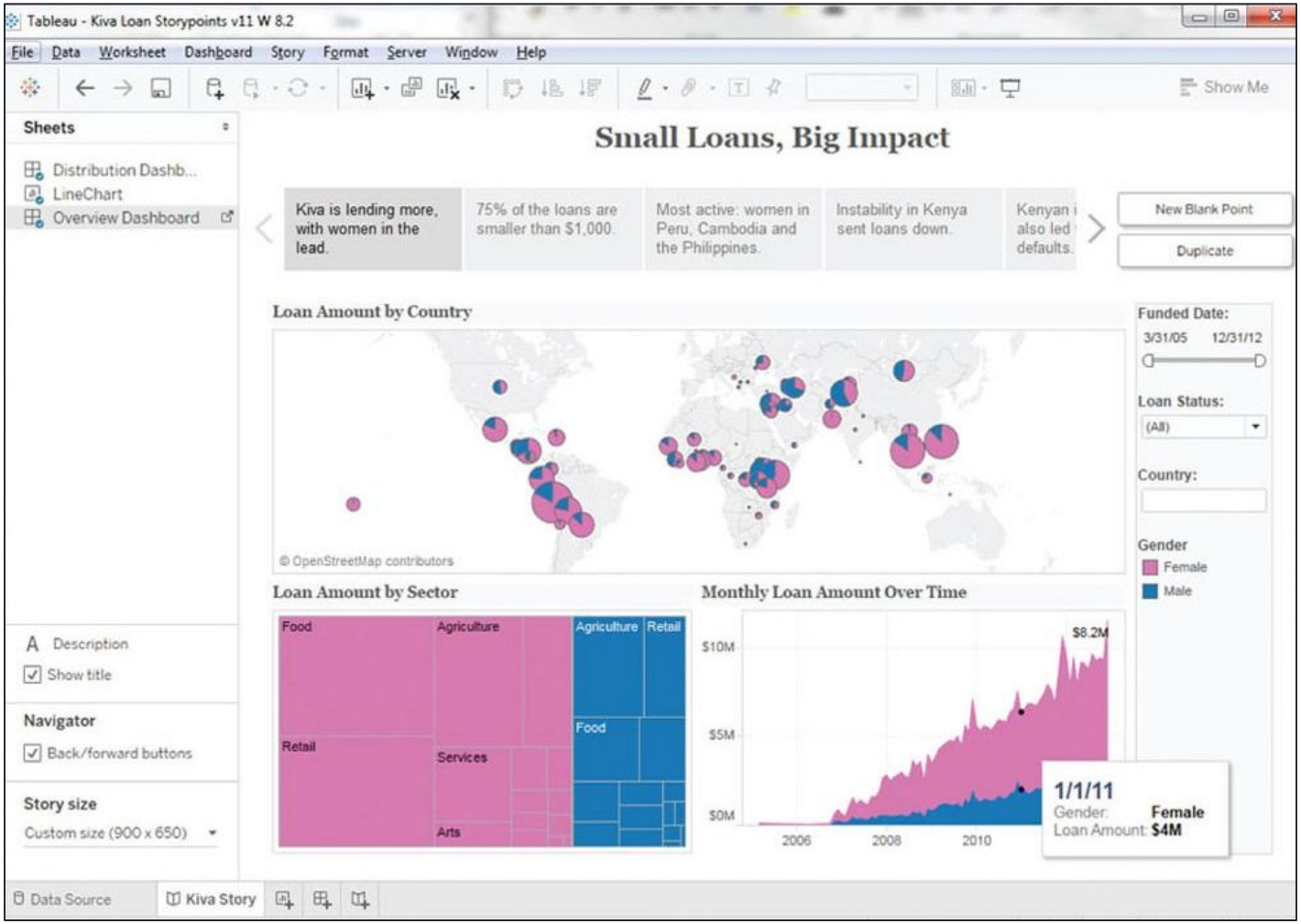
Visual Analytics

- A recently coined term
  - Information visualization + predictive analytics
- Information visualization
  - Descriptive, backward focused
  - “what happened” “what is happening”
- Predictive analytics
  - Predictive, future focused
  - “what will happen” “why will it happen”
- There is a strong move toward visual analytics

Visual Analytics in SAS



Storytelling in Tableau



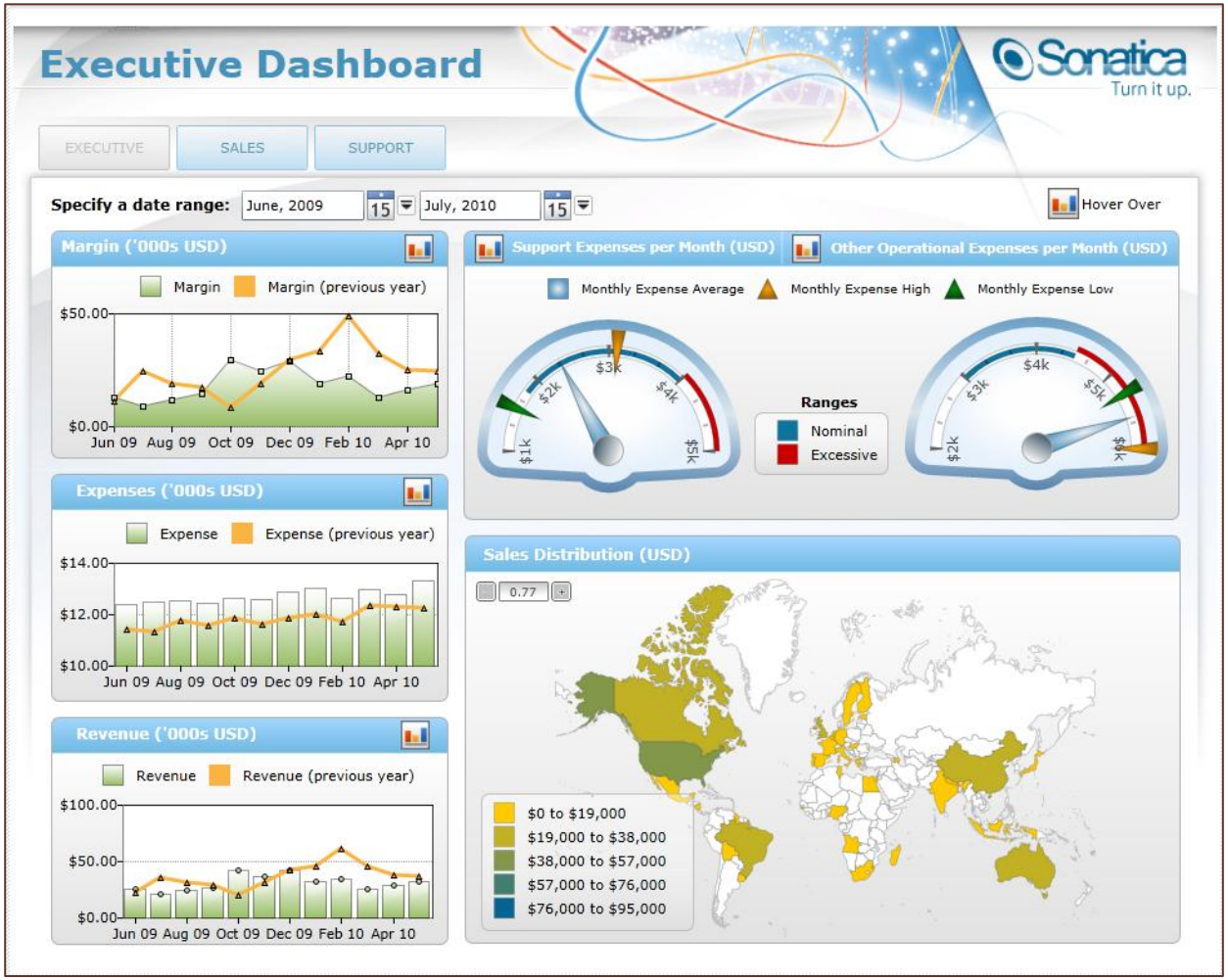
Performance Dashboards

- Performance dashboards are commonly used in BI software
- Dashboards provide visual displays of important information that is consolidated and arranged on a single screen so that information can be digested at a single glance and easily drilled in and further explored

Performance Dashboards

- Dashboard design
  - The fundamental challenge of dashboard design is to display all the required information on a single screen, clearly and without distraction, in a manner that can be assimilated quickly
- Three layer of information
  - Monitoring
  - Analysis
  - Management

Performance Dashboards



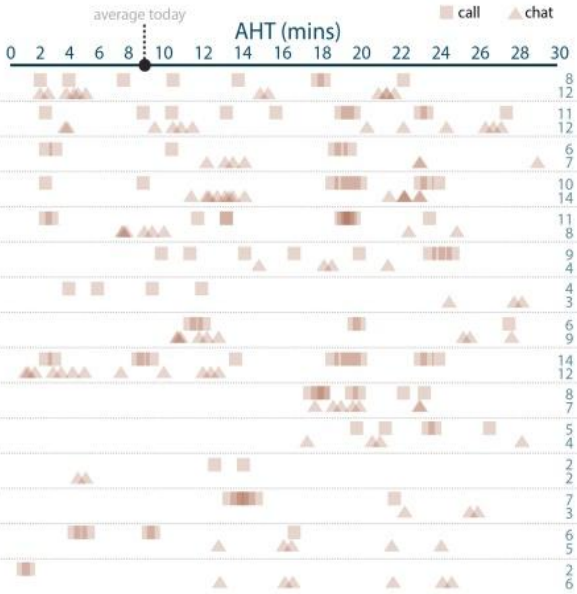
CALL CENTER DASHBOARD

Dundas Data Visualization Inc.

date picker today

built using:

AGENT	FCR(%) 12 wks / this wk	QTD avg: 80%	AHT (mins)	TFRS	CALLS	RATE
Mark Wevers		92		17	32	92
Hermione Granger		89		16	28	89
Ron Weasley		81		12	29	81
Albus Dumbledore		81		11	31	81
Severus Snape		81		18	24	81
Miranda July		76		17	27	76
Sylvia Plath		74		20	22	74
Billy Collins		74		15	25	74
Pablo Neruda		70		11	24	70
Elizabeth Brewster		68		17	22	68
Kurt Vonnegut		68		16	21	68
Richard Brautigan		67		15	18	67
Carl Sagan		65		10	15	65
Colin Firth		62		11	16	62
David Shrigley		61		8	14	61



AGENT	OCCUP (%) target: 78	TOTAL (#)
Snape		24
July		22
Vonnegut		17
Dumbledore		17
Wevers		17
Granger		15
Firth		14
Collins		14
Neruda		12
Brewster		11
Weasley		11
Brautigan		9
Sagan		9
Plath		6
Shrigley		5

inquiry

BILLING

COMPUCOR

AVG SPEED OF ANSWER (secs)

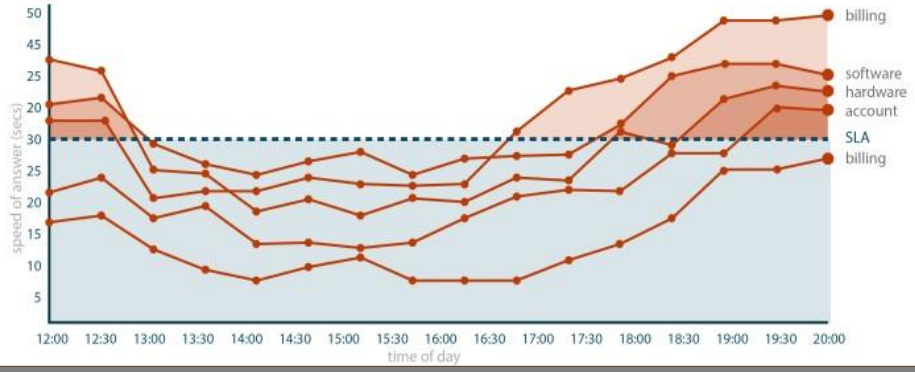
average today
SPEED OF ANSWER
89%<30s

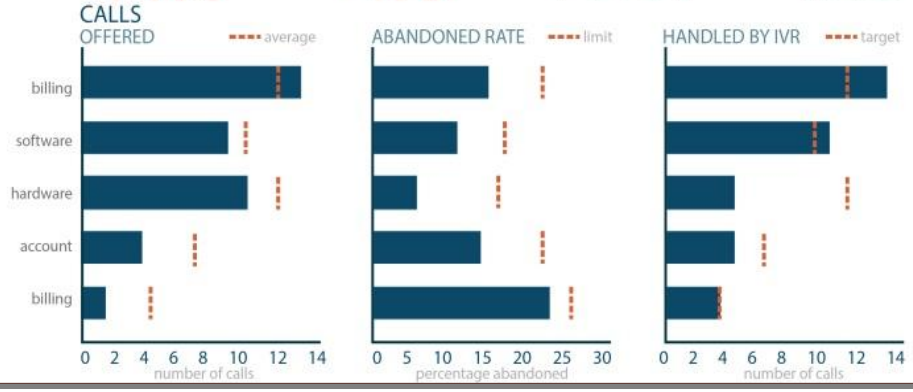
average today
TRANSFER RATE
11%

average today
ABANDON RATE
4%

TOTAL CALLS TODAY
7921

average today
CALLS PER MIN
128





Performance Dashboards

- What to look for in a performance dashboard
  - Use of visual components to highlight data and exceptions that require action.
  - Transparent to the user, meaning that they require minimal training and are extremely easy to use
  - Combine data from a variety of systems into a single, summarized, unified view of the business
  - Enable drill-down or drill-through to underlying data sources or reports
  - Present a dynamic, real-world view with timely data
  - Require little coding to implement, deploy, and maintain

Best Practice in Dashboard Design

- Benchmark the metrics
- Wrap the metrics with contextual metadata
- Validate the design by a usability specialist
- Prioritize and rank alerts and exceptions
- Enrich dashboard with business-user comments
- Present information in three different levels
- Pick the right visual constructs
- Provide for guided analytics

Key Performance Indicators

- **Key performance indicator (KPI)**

  A KPI represents a strategic objective and metrics that measure performance against a goal
- Unique features of KPIs
  - Strategy
  - Targets
  - Ranges
  - Encodings
  - Time frames
  - Benchmarks

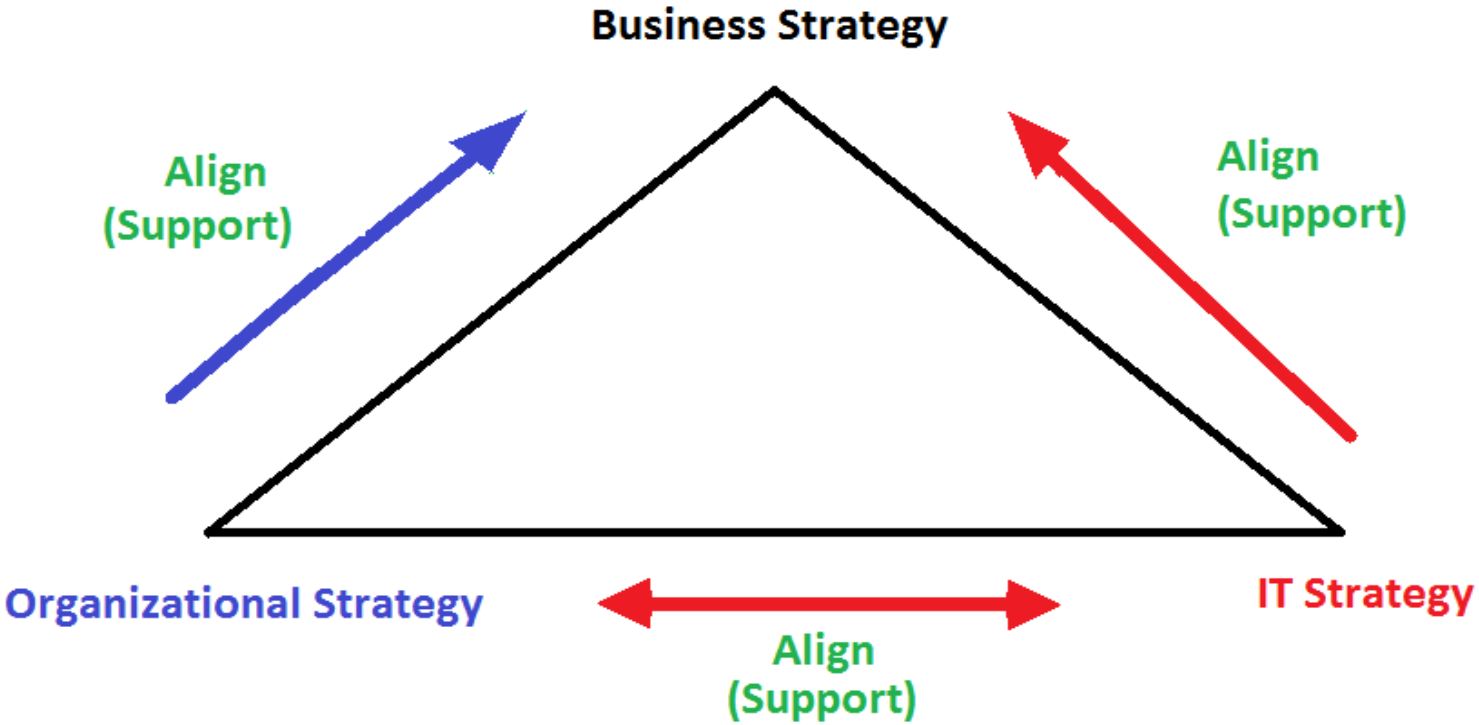
Key Performance Indicators

- **Key performance indicator (KPI)**

Outcome KPIs	vs.	Driver KPIs
(lagging indicators e.g., revenues)		(leading indicators e.g., sales leads)

- Operational areas covered by driver KPIs
  - Customer performance
  - Service performance
  - Sales operations
  - Sales plan/forecast

The Business Strategy Triangle



What is a "Strategy?"

- Coordinated set of actions to fulfill objectives, purposes, or goals
- It sets limits on what the organization seeks to accomplish
- Starts with a mission

Company	Mission Statement
Zappos	To provide the best customer service possible. Internally we call this our WOW philosophy.
Amazon	We seek to be Earth's most customer-centric company for three primary customer sets: consumer customers, seller customers and developer customers.
L.L. Bean	Sell good merchandise at a reasonable profit, treat your customers like human beings and they will always come back for more.

What is a business strategy?

- It is where a business seeks to go and how it expects to get there
- It is not a business model, although it includes business models as one component of a business strategy
- Business models include subscriptions, advertising, licenses, etc.
- Business models do not include where the business seeks to go, and only the revenue portion of how it expects to get there

Generic Strategies Framework

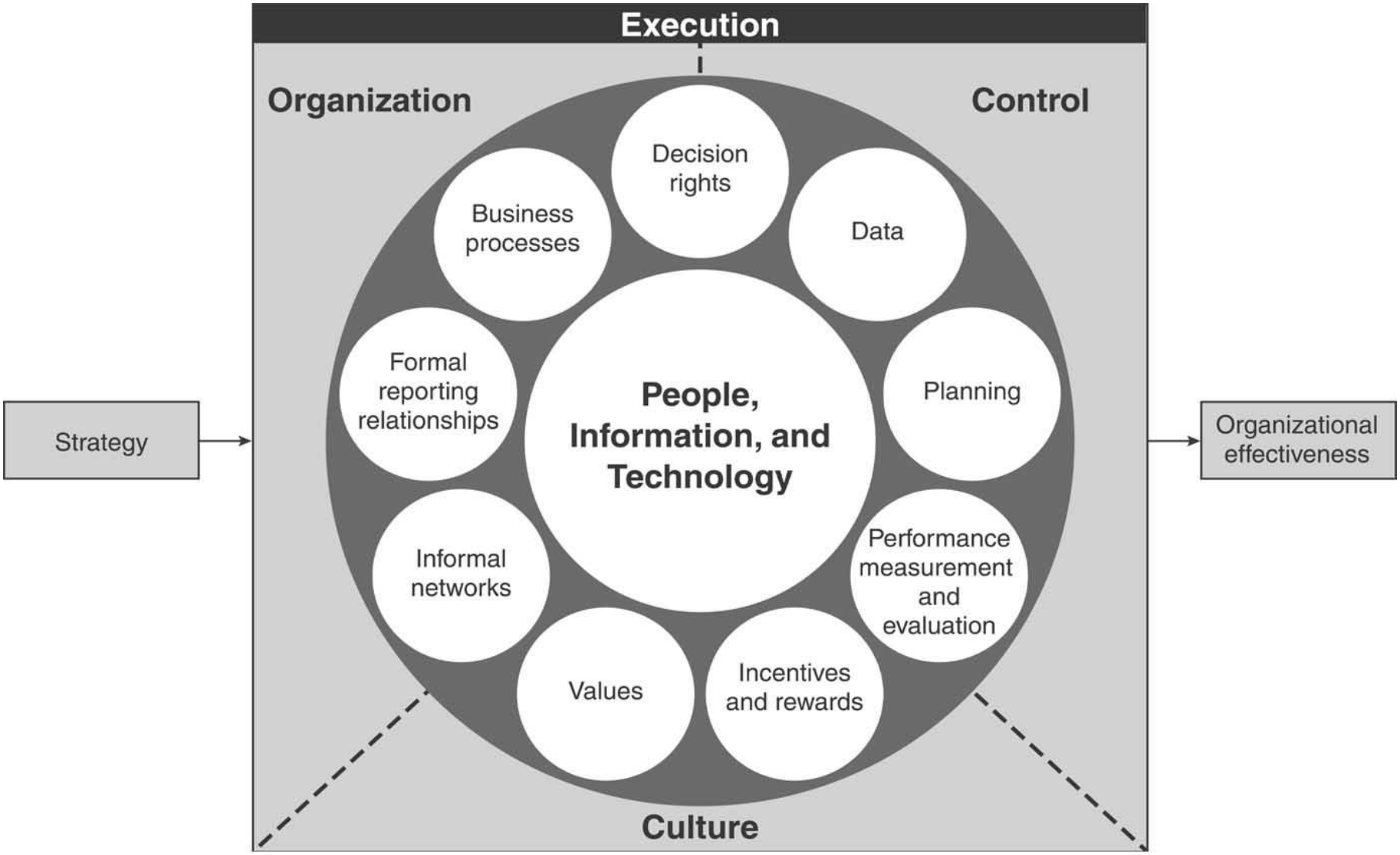
- Michael Porter: How businesses can build a competitive advantage
- Three primary strategies for achieving competitive advantages:
  - Cost leadership – lowest-cost producer.
  - Differentiation – product is unique.
  - Focus – limited scope – can accomplish this via cost leadership or differentiation within the segment

Three Primary Strategies

	Strategic Advantages		
Strategic Target		Uniqueness Perceived by Customer	Low Cost Position
	Industry Wide	Differentiation Apple	Cost Leadership Wal-Mart
	Particular Segment Only	Focus Ritz Carlton	Marriott

Organizational Strategy

- What is organizational strategy?
  - Organizational design and
  - Choices about work processes
- How do you manage organizational, control, and cultural variables?
  - Managerial Levers model
    - Organizational, control and culture variables are managerial levers used by decision maker to effect changes in the organization



Managerial Levers Model

IT Strategies

- What is an IT Strategy?
  - The plan an organization uses in providing information services
- Four key IT infrastructure components

	What	Who	Where
Hardware	The physical devices of the system	System users and managers	Physical location of devices (cloud, datacenter, etc.)
Software	The programs, applications, and utilities	System users and managers	The hardware it resides on and physical location of that hardware
Networking	The way hardware is connected to other hardware, to the Internet and to other outside networks.	System users and managers; company that provides the service	Where the nodes, wires, and other transport media are
Data	Bits of information stored in the system	Owners of data; data administrators	Where the information resides

Social Strategy

- Collaboration
  - Extend the reach of stakeholders to find and connect with one-another
- Engagement
  - Involve stakeholders in the business via blogs; communities
- Innovation
  - Identify, describe, prioritize new ideas

IT Assets and Capabilities

- Information resources as strategic tools
  - IT assets and capabilities (not just assets!)

IT Assets

- IT infrastructure
  - Hardware
  - Software and company apps
  - Network
  - Data
  - Website
- Information Repository
  - Customer information
  - Employee information
  - Marketplace information
  - Vendor information

IT Capabilities

- Technical Skills
  - Proficiency in systems analysis
  - Programming and web design skills
  - Data analysis/data scientist skills
  - Network design and implementation skills
- IT Management Skills
  - Business process knowledge
  - Ability to evaluate technology options
  - Project management skills
  - Envisioning innovative IT solutions
- Relationship Skills
  - Spanning skills such as business-IT relationship management
  - External skills such as vendor management

IT Assets

- Infrastructure
  - Information, technology, people and processes available to perform business processes and tasks (not just technology!)
  - Might even include resources not owned by the firm (e.g., eBay, Facebook, LinkedIn)
- Information repository
  - Data captured, organized, and retrievable by the firm

IT Capabilities

- Technical Skills (designing, developing, implementing IS)
- Management Skills (managing the IT function and IT projects)
- Relationship Skills
  - Outside the organization (vendors, customers)
  - Within the organization (managers)

Sustainability

- Attaining competitive advantages is one thing
  - A firm often exploits rare and valuable resources
- Sustaining competitive advantages is another thing
  - Resources must be difficult to transfer or relatively immobile

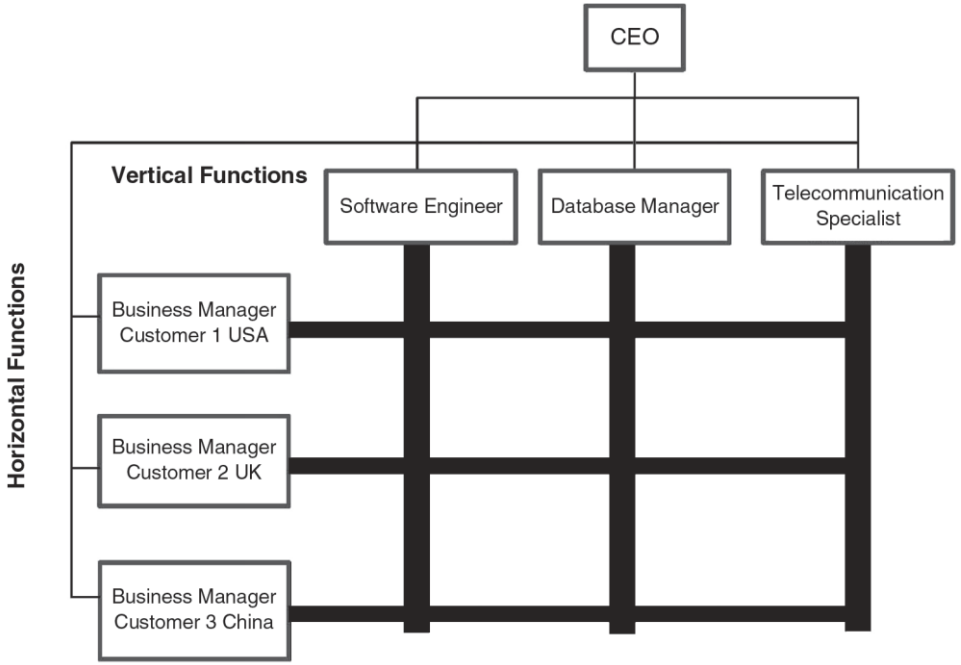
The Resource Based View

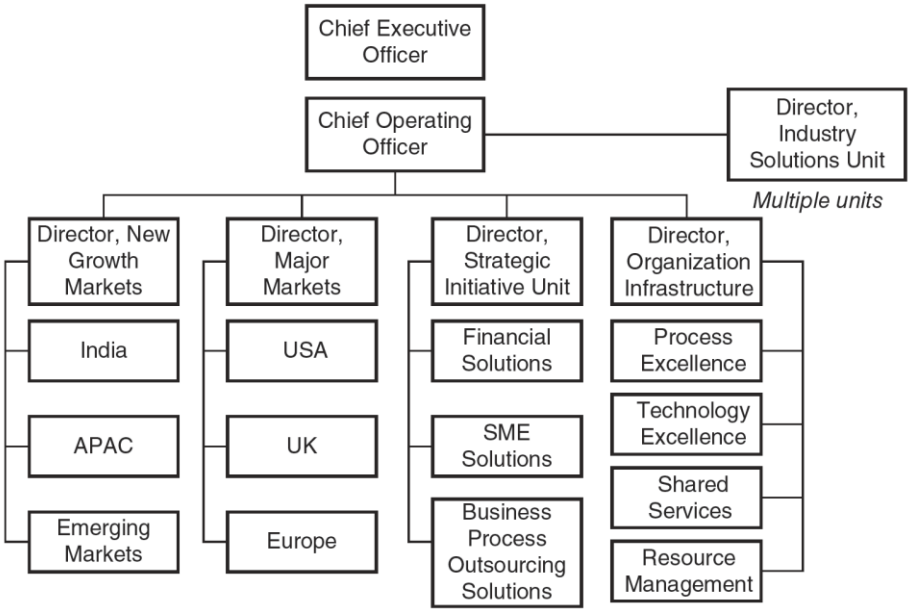
- Resource-based view (RBV)
  - A managerial framework used to determine the strategic resources with the potential to deliver comparative advantage to an organization
  - Involving all information resources (IT assets & IT capacities)

CTS vs TCS

- Both companies are IT consulting/outsourcing firms but have different business strategies
- Cognizant Technology Solutions (CTS)
  - Currently focusing on complex relationship-based solutions
  - The goal was “extremely close partnerships”, needed to be responsive enough to customers
  - Main problem: lack of necessary information flows between software engineers and client customer service managers
  - Adopted a matrix structure to allow functional managers to interact with business managers
- Tata Consultancy Services (TCS)
  - Current focusing on customer and revenue growth
  - The goal was to be nimble and agile
  - Adopted decentralized structure heavy on markets and strategic initiatives

CTS vs TCS





Cognizant

Tata

Variable	Description
Organizational variables	
Decision rights	Authority to initiate, approve, implement, and control decisions necessary to plan and run the business
Business processes	Ordered tasks to complete key business objectives
Formal reporting relationships	Structure set up to coordinate organizational units
Informal networks	Mechanism, such as ad hoc groups, to coordinate and transfer information outside formal reporting relationships.
Control variables	
Data	Facts collected, stored, and used by the organization
Planning	Processes by which future direction is established, communicated, and implemented
Performance measurement and evaluation	Measures to assess successful execution of plans then using the measures to improve the quality of work
Incentives	Monetary and non-monetary devices to motivate behavior
Cultural variables	
Values	Implicit and explicit beliefs that underlie decisions and actions
Locus	The span of culture: local, national, regional, etc.

Organizational Design

- Different designs accomplish different goals
- Decision rights will differ according to the design
- Different reporting relationships and organizational structure will allocate decision rights in different ways
- Four major organizational designs are:
  - Hierarchical
  - Flat
  - Matrix
  - Networked

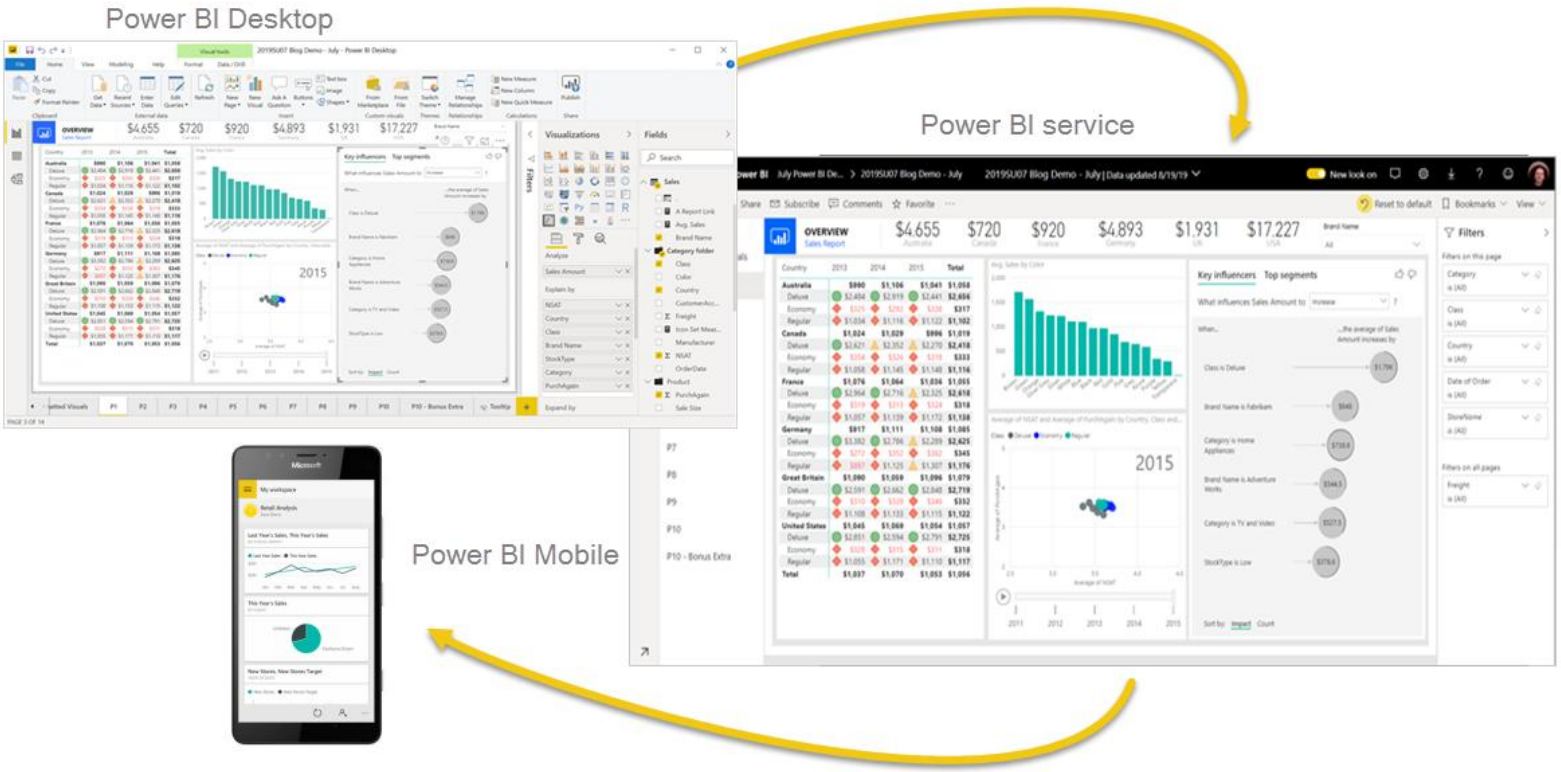
Comparison of Organizational Structures

	Hierarchical	Flat	Matrix	Networked
Description	Bureaucratic w/ defined levels of management	Decision-making pushed down to lowest level	Workers assigned to 2 or more supervisors	Formal/informal communication networks that connect all
Characteristics	Division of labor specialization, unity of command	Informal roles; often small, young organizations	Dual reporting based on function/purpose	Known for flexibility and adaptability
Type of Environment Best Supported	Stable Certain	Dynamic Uncertain	Dynamic Uncertain	Dynamic Uncertain
Basis of Structuring	Primary functional	Very loose	Functions and purpose	Networks
Power Structure	Centralized	Decentralized	Distributed	Distributed
Key Tech. Supporting this	Mainframe, centralized data and processing	PCs	Networks	Intranets and Internet

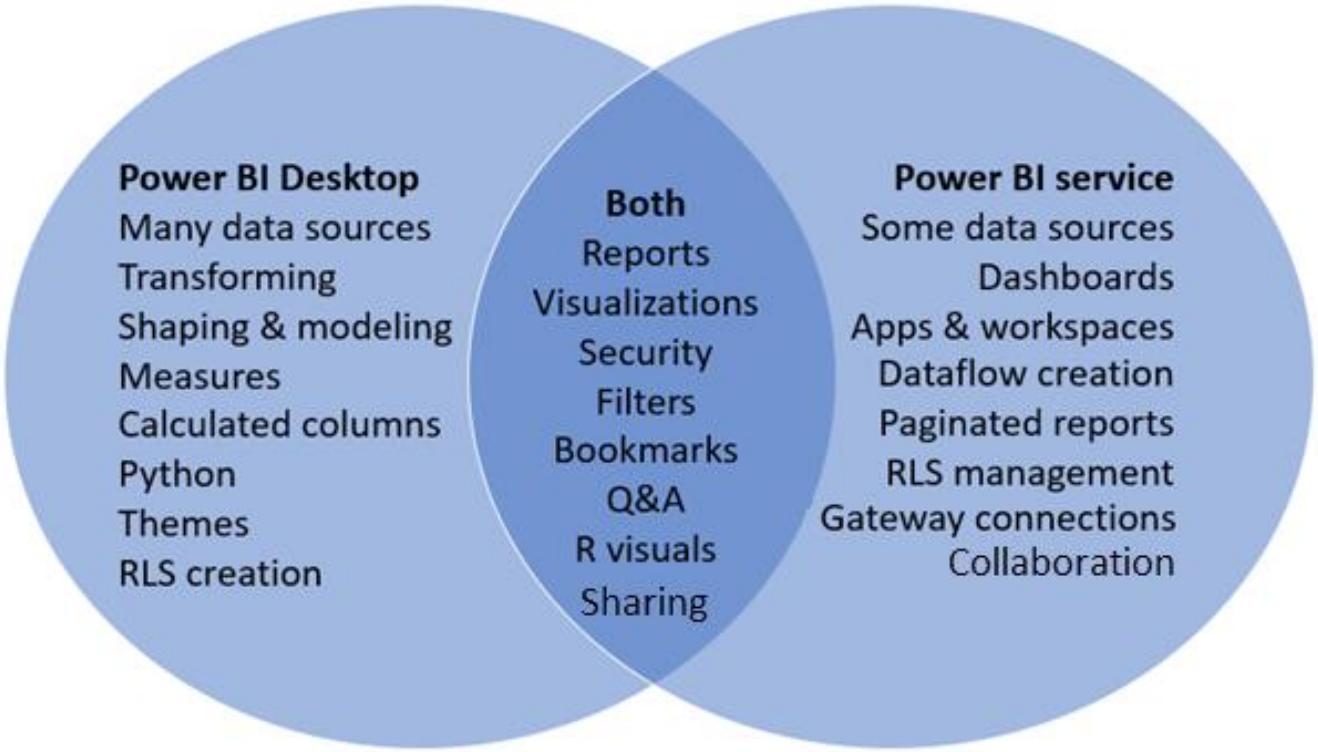
New Options

- Social networks for
  - Finding experts
  - Getting to know colleagues
  - Seeing who has relevant experience for projects across functions and geography
- Virtual teams – geographically and/or organizationally dispersed coworkers
  - Assembled using telecommunications and IT
  - The aim is to accomplish an organizational task

Microsoft Power BI Products



Power BI Desktop vs Power BI Service



Power BI Licensing Models

- Power BI Desktop and Power BI Service Fabric Free
  - Supporting single users
- Power BI Pro
  - For Power BI Service
  - Sharing data, dashboards, reports, etc., among multiple users for team collaboration
- Power BI Premium
  - For Power BI Service
  - Supporting dedicated Power BI Report Servers
  - 60-day free trial

    https://powerbi.microsoft.com/en-us/get-started/

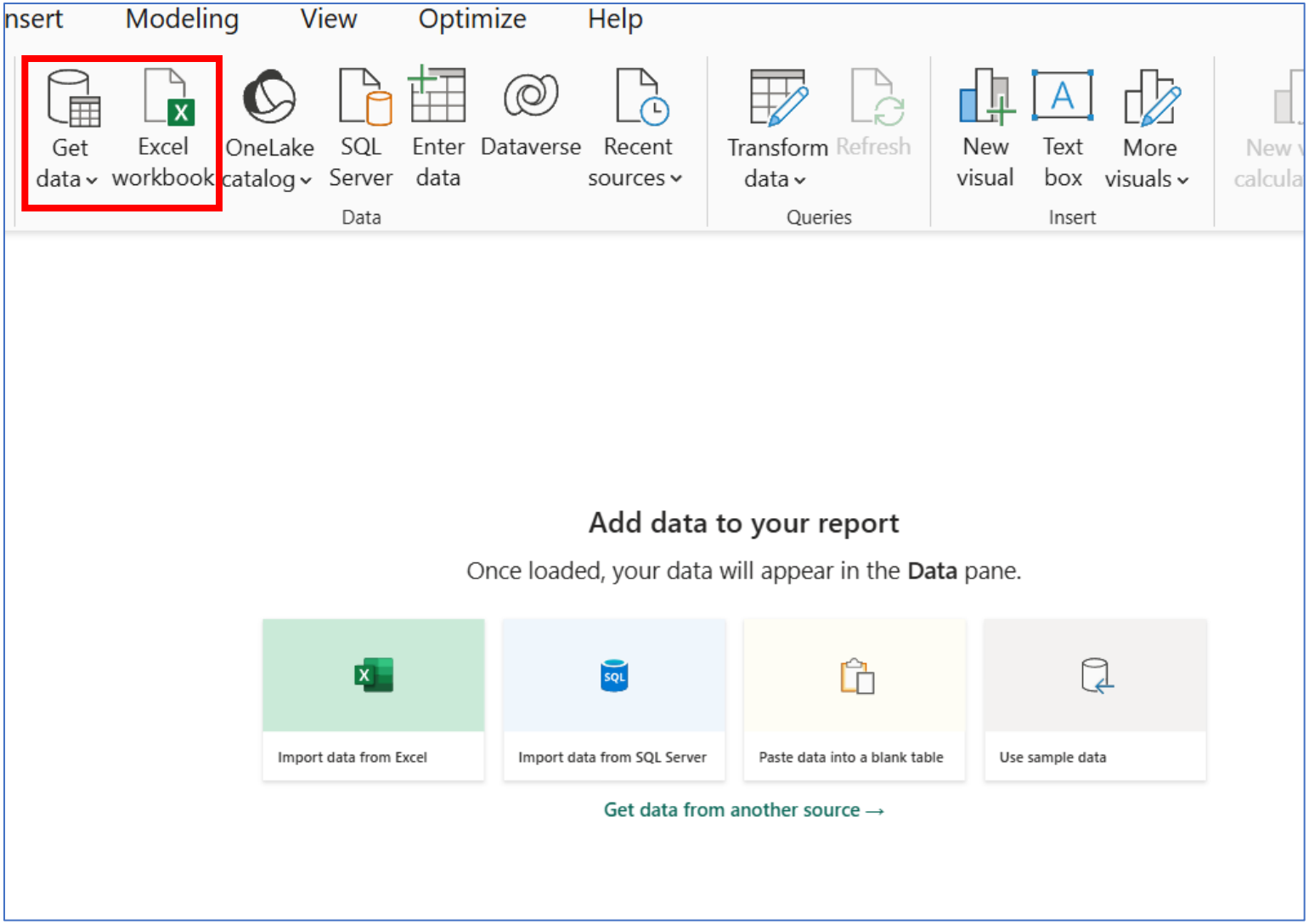
Power BI Objects

- Visuals
  - A visualization (or visual) is a type of chart for displaying the data from datasets
  - Visuals are interactive
  - A *tile* is a rectangular box that holds an individual visual
- Datasets
  - A dataset is a container of data
- Dashboards
  - A dashboard is a single screen with visuals and text to facilitate business decision-making
  - Generally, a dashboard addresses a central area of interest or answers a single question
  - A dashboard can be based on multiple datasets
    - Current limitations: the interactive visuals in dashboards may be much slower than those in reports
  - Current limitation: Power BI Desktop does not support dashboards
- Reports
  - A report is one or more pages of visuals and text
  - A report is based only on a single dataset
    - Multiple data sources must be combined into a single dataset first
  - Generally, each page in a report addresses a central area of interest or answers a single question

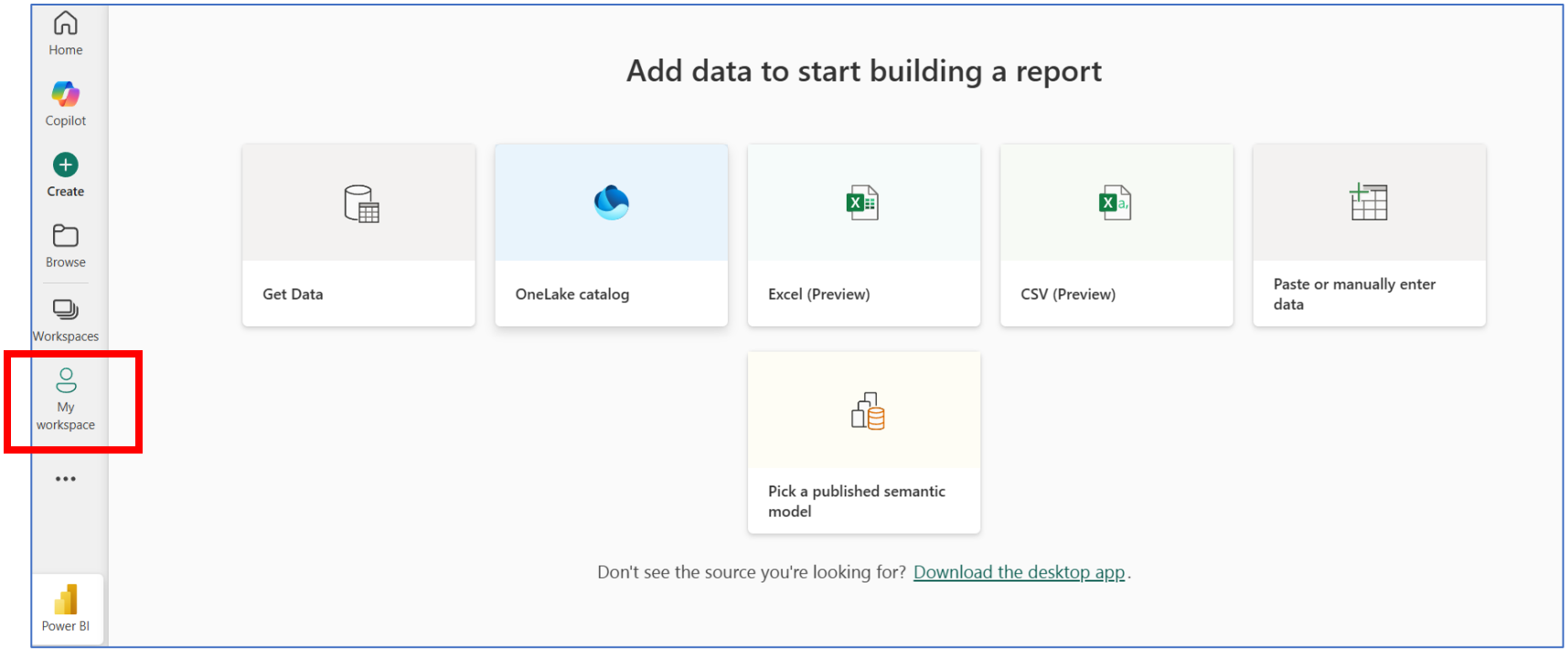
Power BI User Roles

- Business Users
- Dashboard/Report Creators
- Administrators
- Advanced Developers
  - Creating own types of visuals
  - Creating Power BI embedded analytics solutions (i.e., embedding Power BI objects in a web application)
  - Using the Power BI API in other software
  - Integrating Power BI with other analytics tools

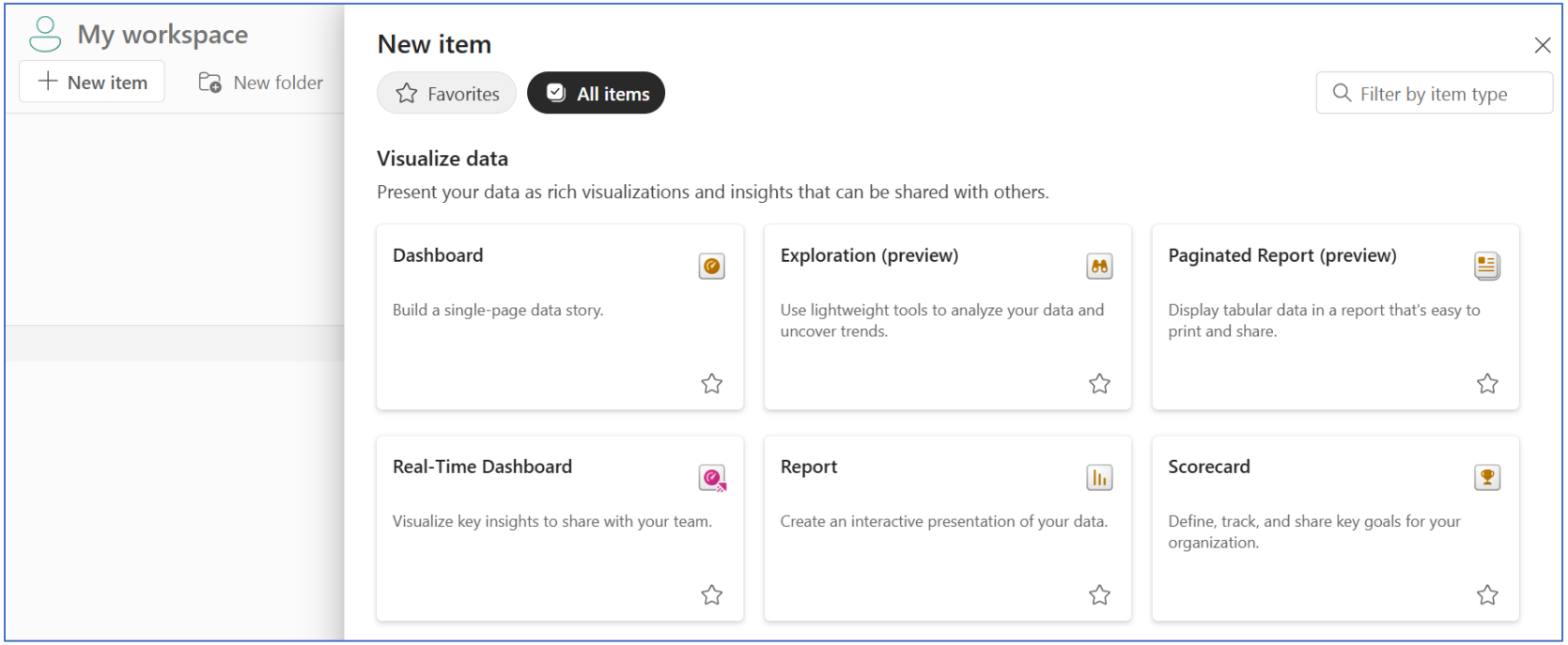
Data Sources on Power BI Desktop



Data Sources on Power BI Service



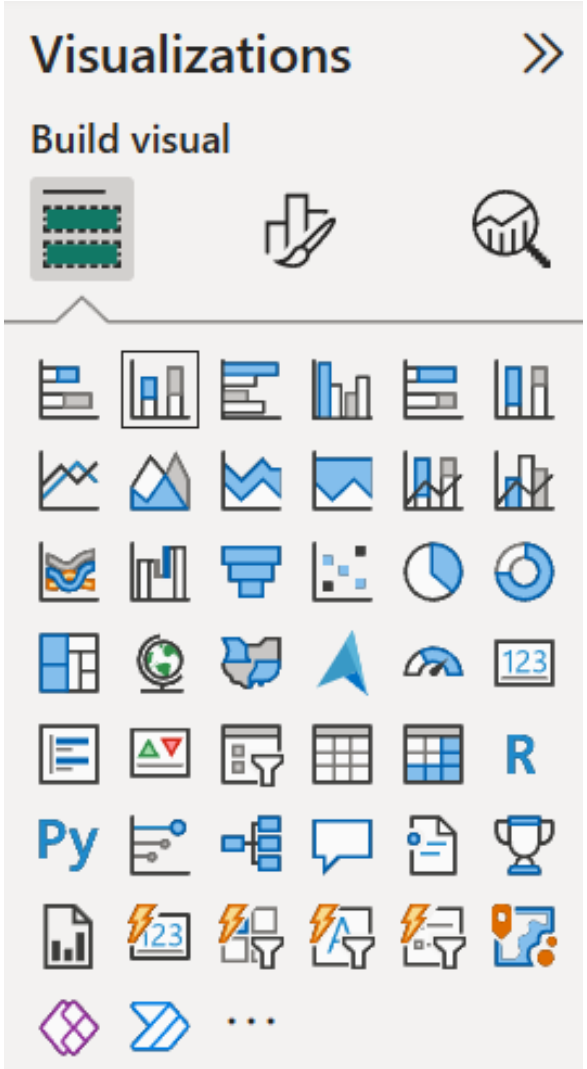
Creating a Report in Power BI Service



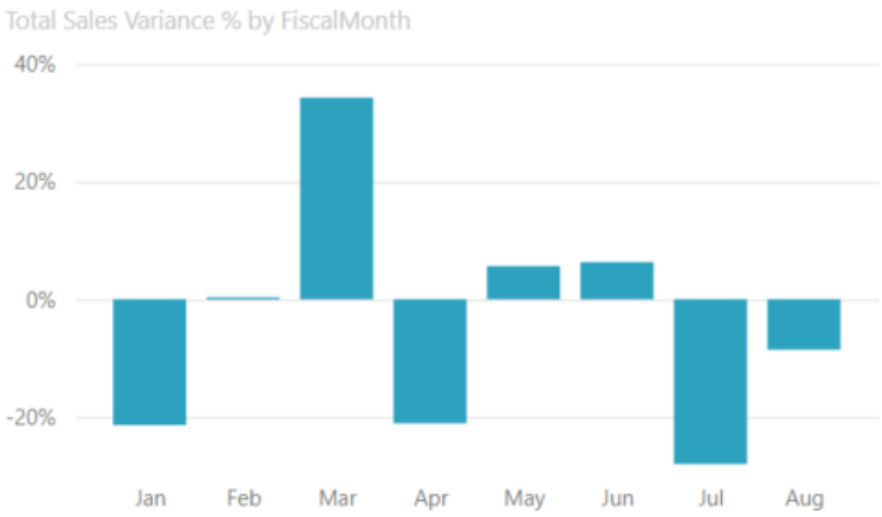
Two Modes of Reports in Power BI Service

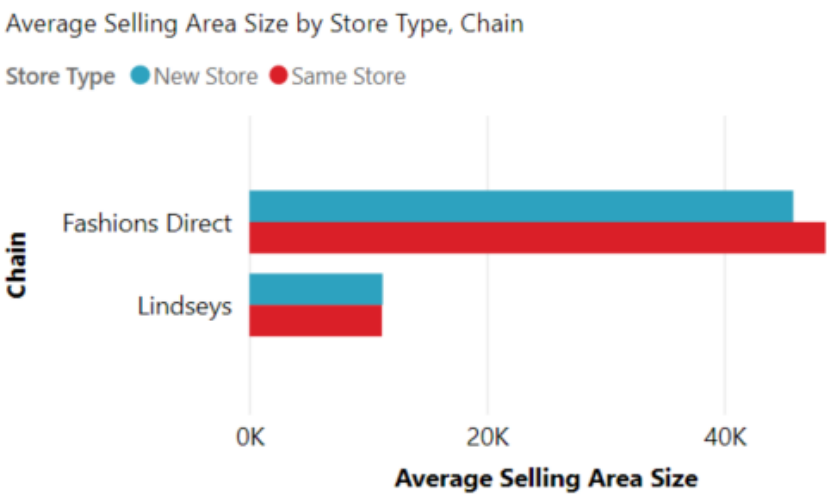
- Two modes for interacting with reports
  - Editing view
    - Used by report creators
  - Reading view
    - Used by business users
- If you are the owner of a report, you can freely switch one view mode to the other

Power BI Visuals

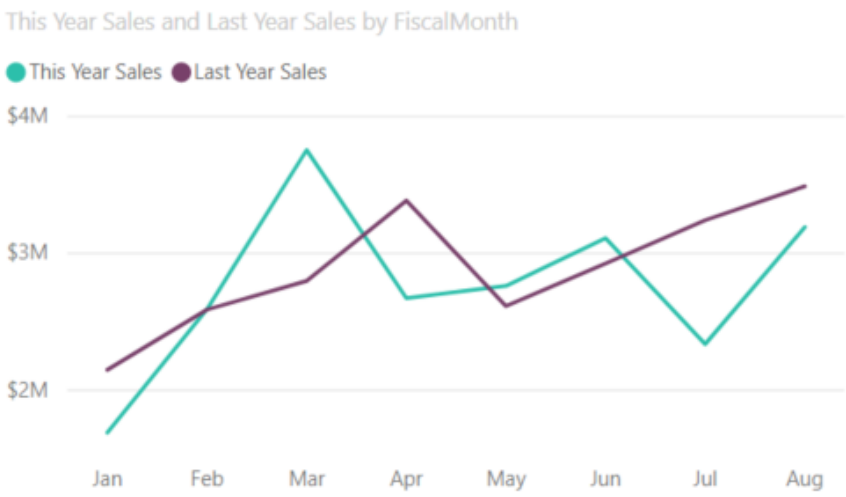


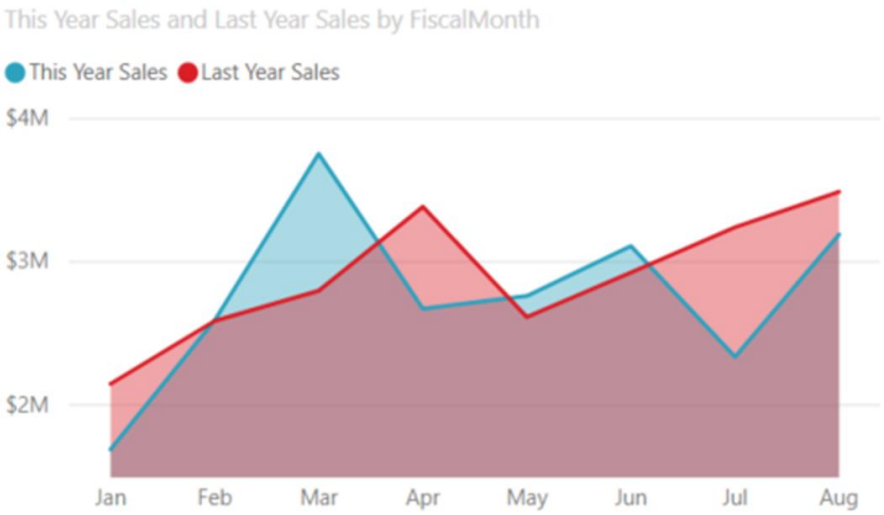
Bar and Column Charts





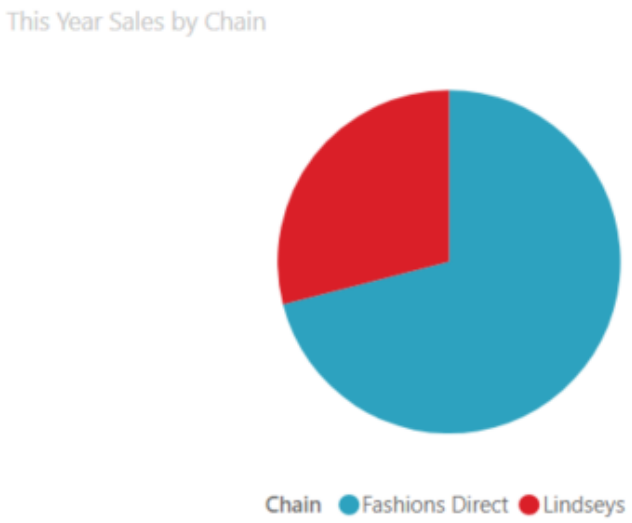
Line and Area Charts

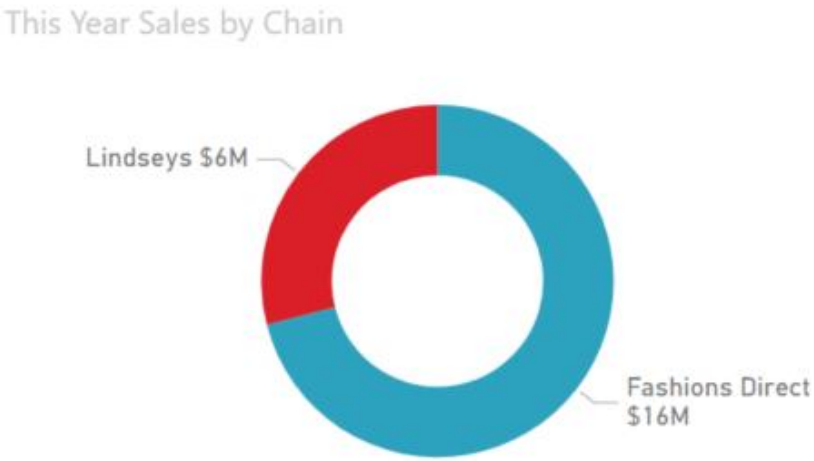




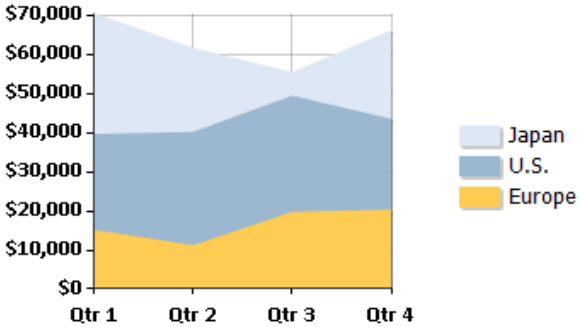
Combo Charts

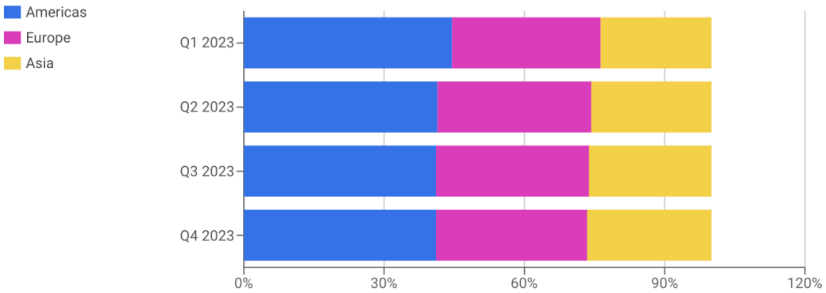
Pie and Donut Charts

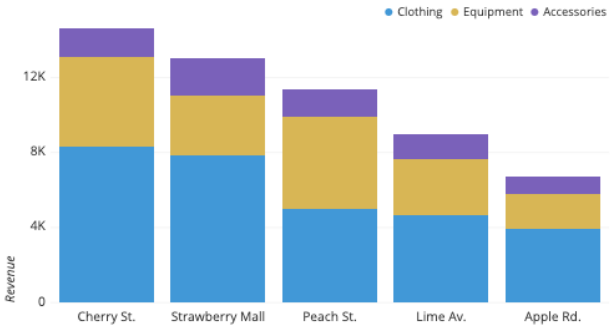


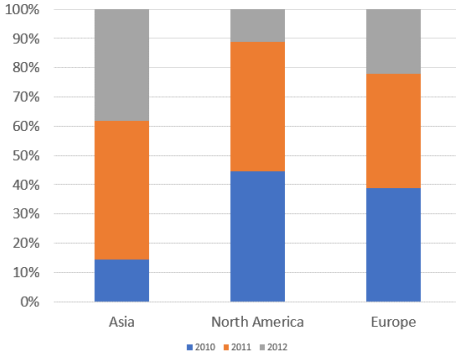


Stacked Charts

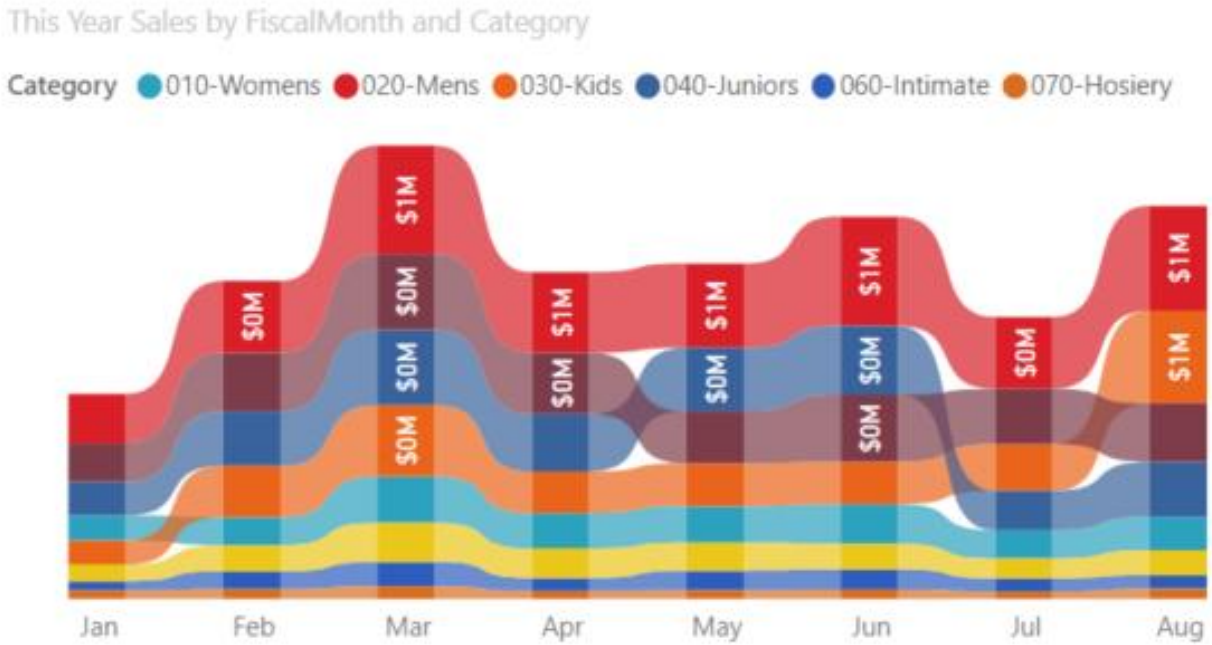




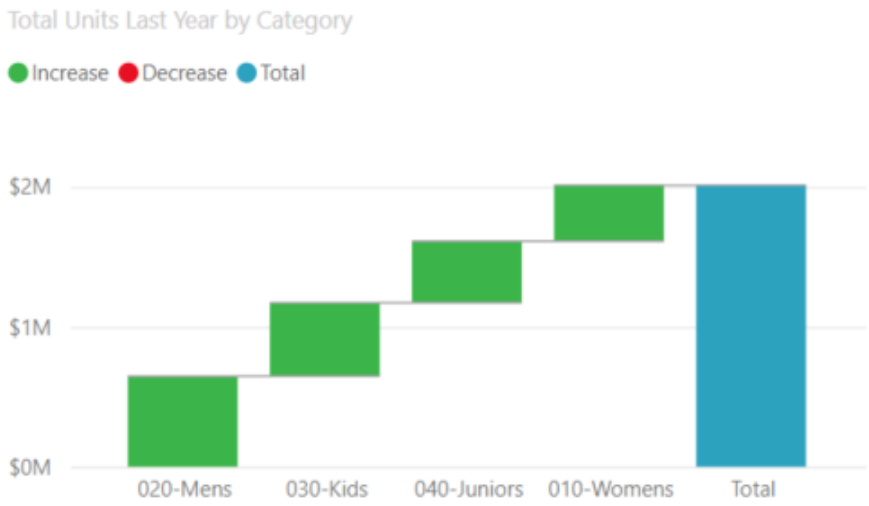




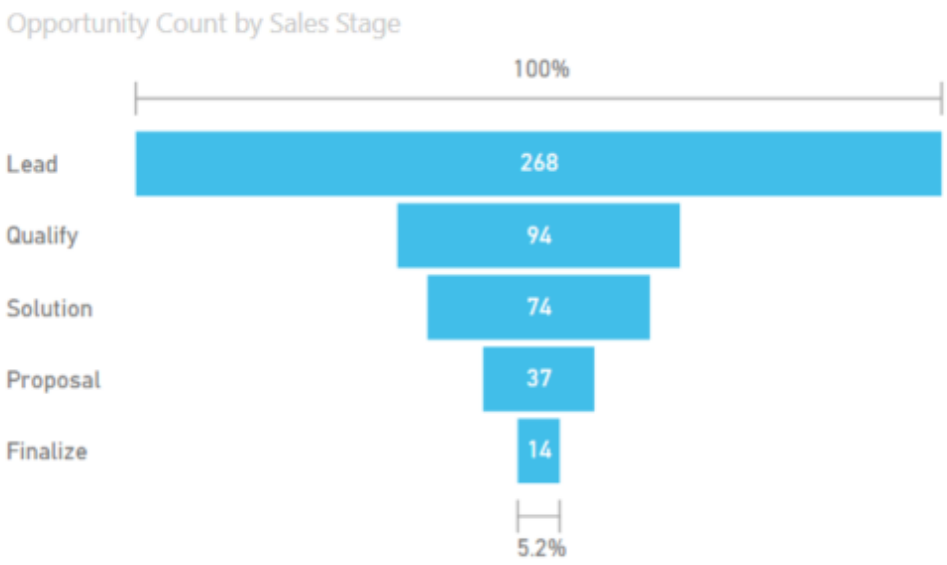
Ribbon Charts



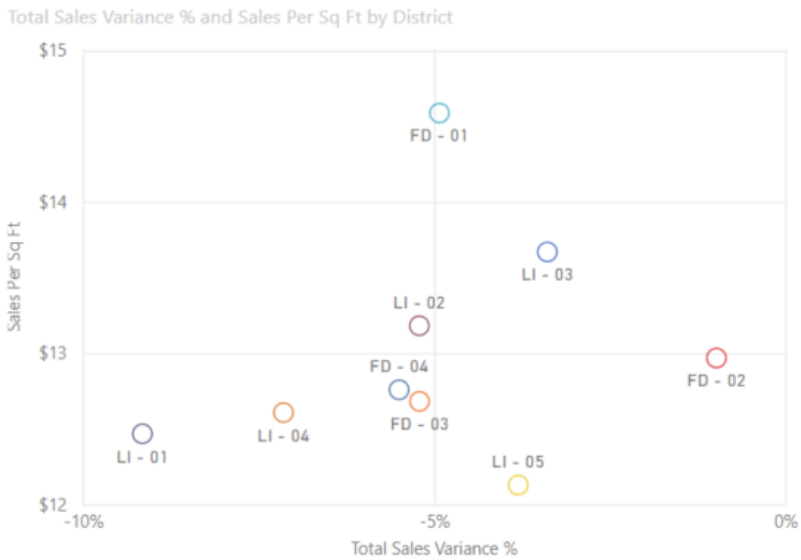
Waterfall Charts

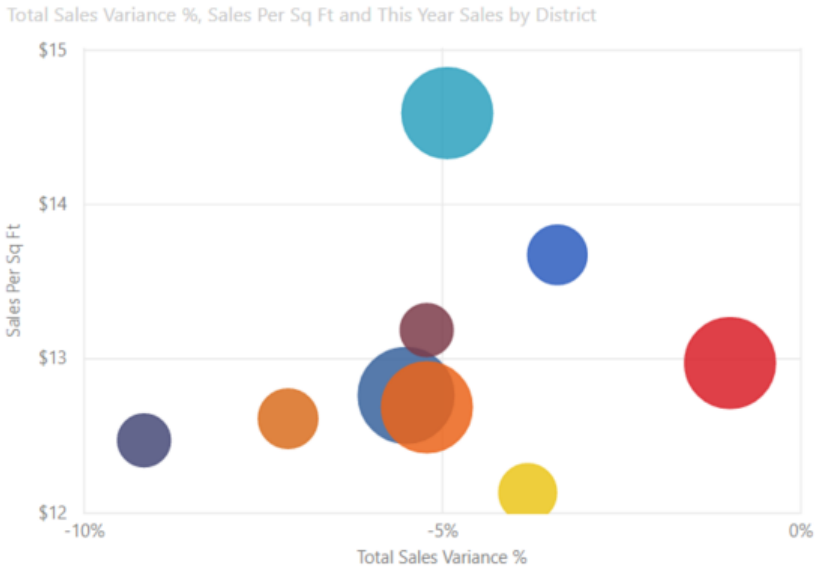


Funnel Charts

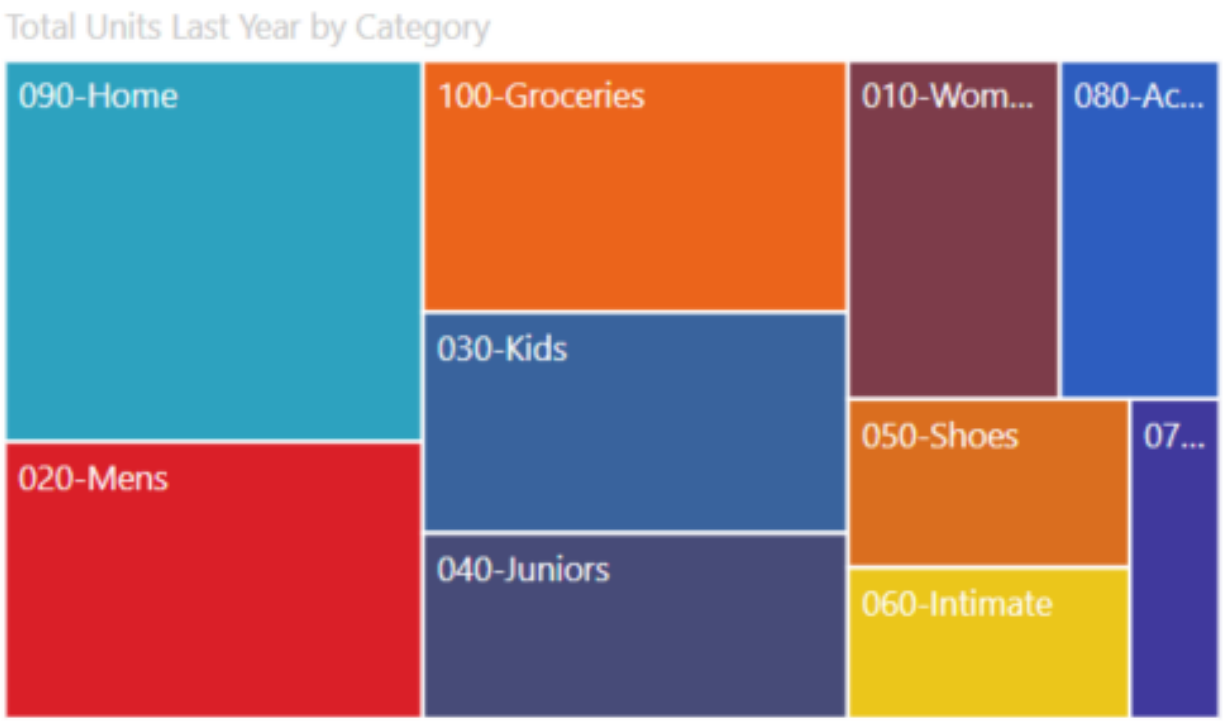


Scatter and Bubble Charts

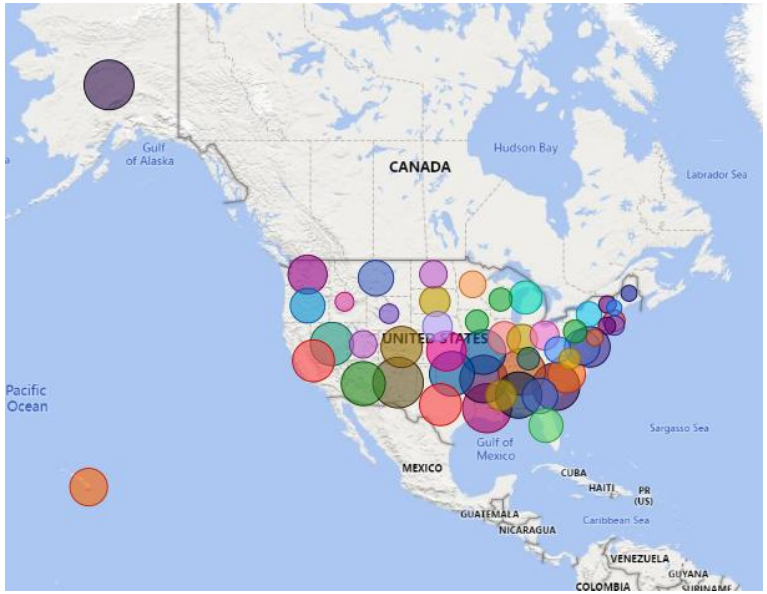


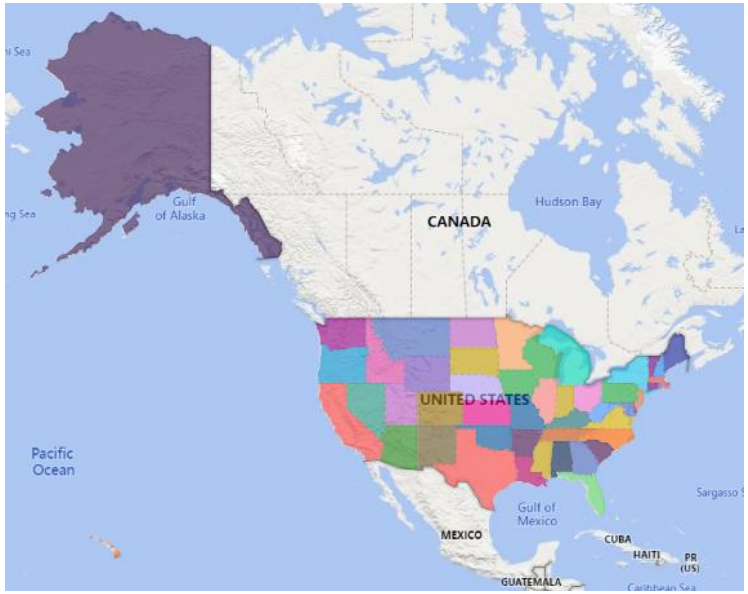


Treemaps

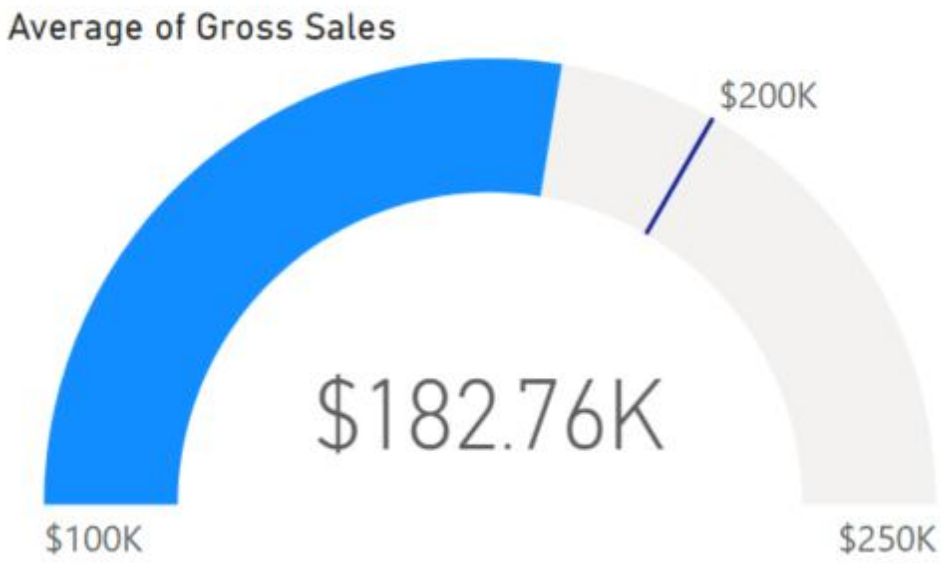

Total Units Last Year by Category
090-Home
100-Groceries
010-Wom...
080-Ac...
030-Kids
050-Shoes
07...
020-Mens
040-Juniors
060-Intimate

Maps and Filled Maps





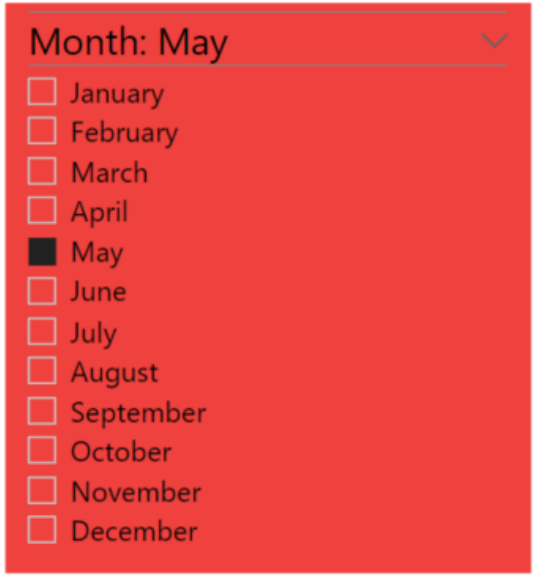
Radial Gauge Charts



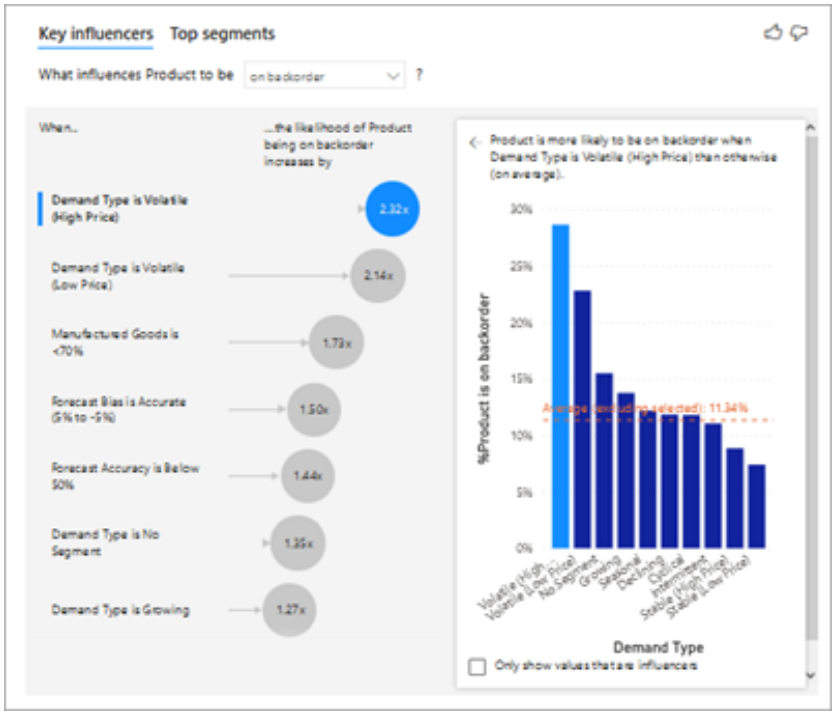
KPI Charts

Total Units This Year and Total Units Last Year by Month
$482,537
Goal: $595,092 (-18.91%)

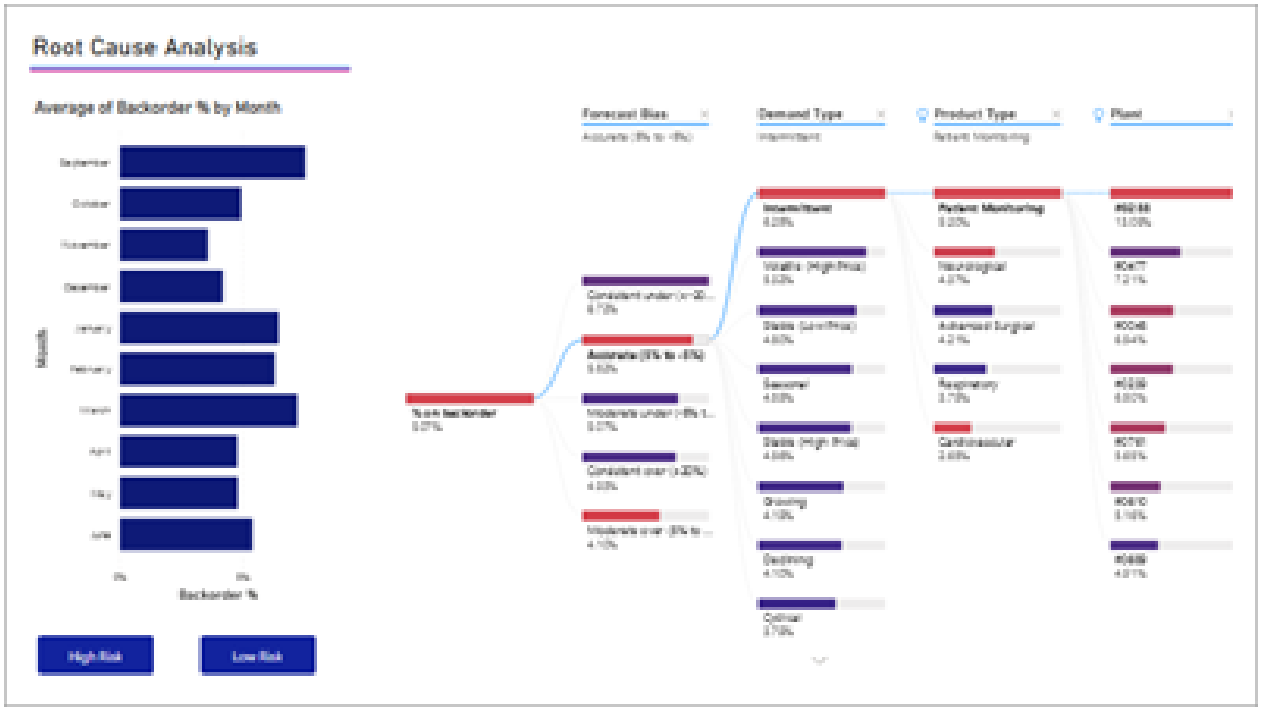
Slicer


Month: May
January
February
March
April
May
June
July
August
September
October
November
December

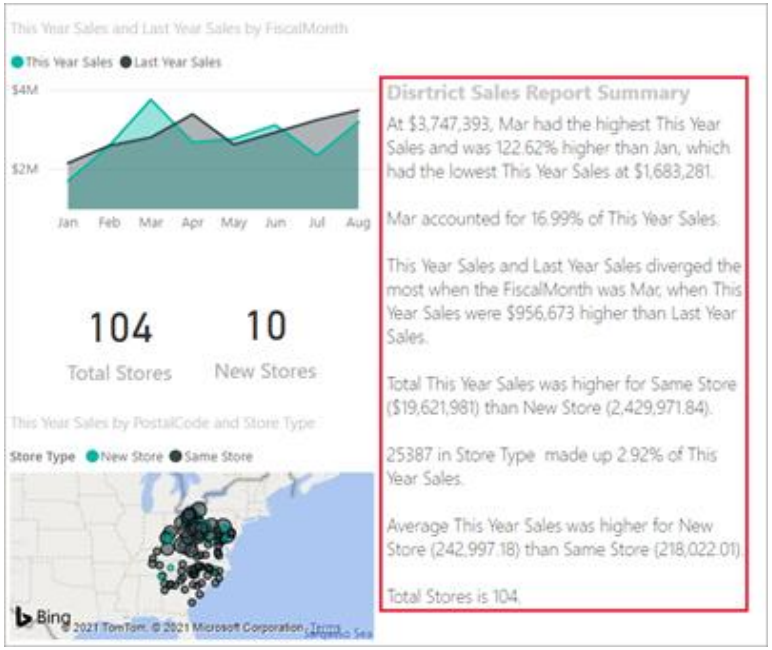
Key Influencers



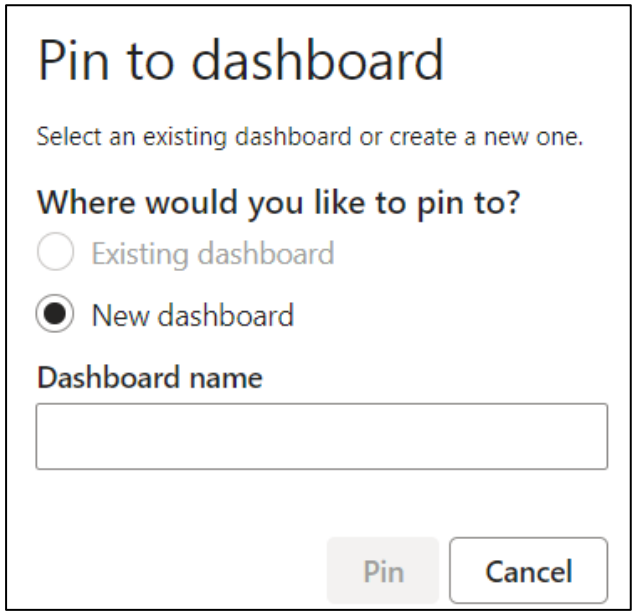
Decomposition Tree



Smart Narrative



Creating a Dashboard from Reports



Report/Dashboard Design

- Report/dashboard design
  - The fundamental challenge of report/dashboard design is to display all the required information clearly and without distraction, in a manner that can be assimilated quickly
- Three layer of information
  - Monitoring
  - Analysis
  - Management

Design Principles

- Use of visuals to highlight data
- Transparent to business users, meaning that they require minimal training and are extremely easy to use
- Combining data from a variety of sources into a single, summarized, and unified view
- Presenting a dynamic, real-world view with timely data
- Requiring little coding or no coding to implement, deploy, and maintain

Key Performance Indicators

- Outcome KPI
  - Lagging indicators, e.g., revenues
- Driver KPI
  - Leading indicators, e.g., sales leads
- Unique features of KPIs
  - Strategy
  - Targets
  - Ranges
  - Encodings
  - Time frames
  - Benchmarks